

PRISM Payments Hub - System Overview & Wire Lifecycle State Model

Wire orchestration platform for all CNB channels and rails (Fedwire, Swift CBPR+, book transfer)

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1. Overview

PRISM Payments Hub (PPH) is CNB's wire orchestration platform, built on the **Volaris Payment Platform 9.4** (vendor) and operated on-premises in an active-active configuration across the Columbus and Charlotte data centers. PPH receives payment instructions from all channels, validates and enriches them, coordinates screening and holds with the Payment Risk & Screening Platform (PRSP), performs funds control against the Core Deposit Platform (CDP), and routes to the Payment Network Gateway (Fedwire Funds Connector or Swift Alliance/gpi Connector).

Dimension	Value (Q2-2026 average per business day)
Outgoing domestic wires (Fedwire)	14,200
Outgoing international wires (Swift CBPR+)	2,900
Incoming wires (all rails)	18,500
Book transfers	6,300
Channels	CBO (~67% of outgoing), host-to-host files, Swift for Corporates (MT101/pain.001), branch, Ops desk
Availability target	99.95% during Fedwire operating hours
Vendor roadmap	Volaris 9.6 upgrade scheduled April 2027 (removes legacy v1 API adapter)

2. Processing stages

1. **Intake** - channel submits instruction (v1 or v2 API, file, Swift).
2. **Validation & enrichment** - format, routing (ABA/BIC), ISO 20022 mapping, structured address checks (mandatory for Swift CBPR+ and Fedwire from November 2026).
3. **Screening** - synchronous call to PRSP (SanctionScreen + Sentinel). A hit creates an HMS hold and the payment enters HELD.
4. **Funds control** - memo debit against CDP; insufficient funds create HRC-05 holds with auto-retry until 5:30 p.m. ET.
5. **Release** - UETR assigned (all outbound wires since ADR-PAY-023), message built (pacs.008 / pacs.009).
6. **Network** - sent to Fedwire Funds Connector or Swift Alliance/gpi Connector; acknowledgment updates state.
7. **Close** - accounting close (COMPLETED); later returns (pacs.004) transition to RETURNED.

3. Canonical lifecycle states (v2)

State	Meaning	Typical duration	v1 status shown to consumers
RECEIVED	Instruction accepted by PPH	< 1 s	RECEIVED
VALIDATING	Format/routing/enrichment	1-5 s	PENDING
SCREENING	PRSP screening in progress	1-20 s	PENDING
HELD	One or more HMS holds open	minutes-days	HELD
REPAIR	Wire Room repair queue	minutes-hours	PENDING

State	Meaning	Typical duration	v1 status shown to consumers
FUNDS_CONTROL	Awaiting available balance	seconds-hours	PENDING
WAREHOUSED	Future-dated or after cutoff	until value date	PENDING
RELEASED	Released for transmission, UETR assigned	seconds	PROCESSED
SENT_TO_NETWORK	Message delivered to network gateway	seconds-minutes	PROCESSED
NETWORK_ACCEPTED	Network acknowledgment received (rail-specific, see 4)	-	PROCESSED
NETWORK_REJECTED	Network rejected (admi.002 / pacs.002 RJCT / NAK)	-	REJECTED
COMPLETED	Internal accounting close	end of day	PROCESSED
CANCELLED	Cancelled before release	-	CANCELLED
RETURNED	Funds returned (pacs.004) after release	hours-days	RETURNED

Note on v1 status PROCESSED

v1 collapses RELEASED, SENT_TO_NETWORK, NETWORK_ACCEPTED and COMPLETED into a single value PROCESSED. Consumers of v1 cannot distinguish a wire that has merely been released from one acknowledged by the network.

4. Rail-specific meaning of NETWORK_ACCEPTED

Rail	Acknowledgment source	Business meaning	Finality
Fedwire Funds	Federal Reserve pacs.002 positive acknowledgment carrying OMAD (via net.fedwire.ack.v1)	Payment order accepted and settled by the Federal Reserve; funds credited to the receiving bank's master account	Final and irrevocable interbank settlement (Regulation J / UCC Article 4A). Does not confirm credit to the beneficiary's account.
Swift CBPR+	SwiftNet delivery acknowledgment (ACK) for pacs.008	Message accepted by the Swift network for delivery to the next agent	No settlement implication. Settlement occurs through correspondent (nostro/vostro) accounts; beneficiary credit is known only from gpi ACCC (if available).

The v2 stateHistory timestamp for NETWORK_ACCEPTED is the time PPH processed the acknowledgment, not the Federal Reserve's creation timestamp in pacs.002. The authoritative Fed timestamp and OMAD are carried on net.fedwire.ack.v1 (see PPH-2207).

5. Identifiers

Identifier	Assigned by	Scope	Available in
pphId	PPH	Internal payment key	v1, v2, events
cboRef	CBO	Channel reference	v2 identifiers.cboRef (if channel = CBO)
IMAD	PPH (Fedwire sender)	Fedwire input message	v1 fedRef; v2 identifiers.imad
OMAD	Federal Reserve	Fedwire output message (acceptance)	v2 identifiers.omad; net.fedwire.ack.v1
UETR	PPH at release	End-to-end tracking id (Swift gpi; carried on Fedwire pacs.008)	v2 identifiers.uetr; lifecycle v2 events
EndToEndId	Originator / PPH	Client reference passed through	v2

6. Integration with Hold Management (PRSP-HMS)

When screening or funds control raises a hold, HMS returns a holdId and PPH transitions the payment to HELD. Since the 2019 integration, PPH also synchronizes the HMS hold *description* into the legacy v1 field **holdReasonDesc** for backward compatibility with the Wire Room console. The v2 API exposes only hold.isHeld and hold.holdId (ADR-PAY-021).

7. Cutoffs and operating windows

Item	Value
Fedwire operating day	9:00 p.m. ET (prior calendar day) to 7:00 p.m. ET
Fedwire third-party customer transfer cutoff	6:00 p.m. ET (later items warehoused, HRC-04)
CBO same-day wire cutoff (domestic)	5:30 p.m. ET

Item	Value
International cutoffs	By currency (e.g., EUR 2:00 p.m. ET, GBP 1:00 p.m. ET, MXN 3:00 p.m. ET, JPY prior day)

8. Upcoming changes

- **November 2026** - structured/hybrid postal address enforcement for Swift CBPR+ and Fedwire; repair holds (HRC-06) expected to rise.
- **FedNow outbound expansion** - PI 27.1 commitment.
- **2027-03-31** - PPH v1 API sunset (PPH-2190); **April 2027** - Volaris 9.6 upgrade.